

Beyond Visual Quality: Fidelity-Oriented Diffusion Model for Real-world Image Super-Resolution

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Abstract

Although existing diffusion-based image super-resolution methods have achieved remarkable visual quality, they often struggle with fidelity issues, particularly in preserving consistency with the original input image. This issue arises because using low-quality images as conditional inputs introduces substantial errors in the diffusion backward denoising process, making the restored features deviate from target features and thus degrade image fidelity. To improve the accuracy of noise estimation, we propose a dual-memory module to reinforce the input low-quality conditional features, which consists of a pre-trained high-quality memory bank to enrich the structural information and a degradation memory to remove the degradation components. Furthermore, we develop an uncertainty-aware noise estimation framework, utilizing an extra branch in the denoising network to predict pixel-wise uncertainty values, thus dynamically adjust the optimization weights for high-uncertainty regions. This adaptive strategy effectively improves the accuracy of noise estimation in challenging reconstruction areas. Experimental results demonstrate that our method significantly enhances the fidelity while preserving high visual quality of diffusion-based super-resolution, improving the reliability of diffusion applications.

CCS Concepts

• **Computing methodologies** → **Reconstruction.**

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3755057>

Keywords

Real-World Image Super-Resolution, Diffusion-based SR, Adaptive Weighted Loss

ACM Reference Format:

Zhenxuan Fang, Shuaibo Wang, Weisheng Dong, Junwei Xu, Fangfang Wu, Xin Li, and Guangming Shi. 2025. Beyond Visual Quality: Fidelity-Oriented Diffusion Model for Real-world Image Super-Resolution. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3746027.3755057>

1 Introduction

Image Super-Resolution (SR) is a crucial vision task focused on restoring a high-resolution (HR) image from its low-resolution (LR) counterpart, which has wide applications, motivating research to enhance both visual quality and content fidelity. Traditional SR approaches, including interpolation-based methods [2, 51] and early deep learning models [8, 22, 26, 49, 50], have achieved impressive progress. However, these methods struggle to capture fine-grained details, especially in real-world scenarios where the degradation process is unknown or more complex than simulated conditions.

This limitation promoted the real-world blind SR problems. Real-ESRGAN [37] uses a high-order degradation model to simulate complex real-world degradation patterns, which effectively produces realistic details. DASR [25] learns degradation patterns directly from real-world images, its degradation-aware module adaptively processes images based on the predicted degradation. FeMaSR [6] represents a significant advancement by using a VQ module to apply feature matching for LR inputs. However, these approaches often introduce unrealistic artifacts due to the instability of adversarial training [42], causing the distortion phenomenon of image.

Recently, Stable Diffusion (SD)-based SR methods have attracted attention. Since LDM [30] achieved remarkable success in image generation, SD-based image restoration methods offer a more flexible approach to handling diverse real-world degradations. These

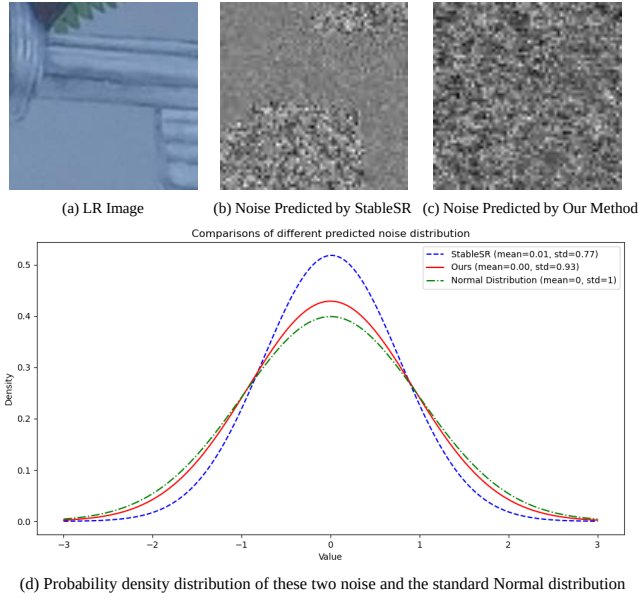


Figure 1: (a) The input LR image. (b) The noise predicted by the diffusion-based SR method StableSR [33] has significant errors, which are non-uniform and entangled with the input image. (c) The noise predicted by our method is more accurate and independent of signals. (d) The probability density distribution of these two noise vs. standard Normal distribution, the noise predicted by our method is closer to the standard Normal distribution.

methods treat the SR task as a generation problem conditioned on the observed LR image, leveraging the powerful generative capabilities of pre-trained SD models to produce highly detailed and visually compelling results. StableSR [33] retains diffusion priors by fine-tuning the time-aware encoder and feature modulation layers. SeeSR [42] uses high-quality semantic prompts to exploit the generative potential of SD models. Despite their capacity for generating realistic textures, SD-based methods often face trade-offs in fidelity, struggling to maintain the structural consistency with the original input image [30, 42]. This limitation prompts us to investigate the underlying reasons for these challenges.

Our analysis of existing diffusion-based SR methods reveals a critical limitation: the noise predicted by the denoising network often becomes entangled with the input image, as illustrated in Figure 1 (b). In principle, the noise should remain statistically independent of the signal, enabling the model to effectively separate the clean latent code from the random noise introduced during the forward diffusion process. However, when the predicted noise correlates with the image content, the reconstructed output exhibits distortions, particularly in regions with complex textures or severe degradation. This issue arises from a common characteristic: these diffusion-based approaches rely on low-quality (LQ) features from the input image as conditional information to guide high-resolution image generation. These LQ features frequently lack essential high-frequency details and structural information, thereby providing inadequate guidance for the generative process.

Moreover, the widely adopted Mean Squared Error (MSE) loss in diffusion models imposes uniform weighting across spatial regions, leading to suboptimal reconstruction in areas that are inherently more challenging to restore. As a result, the stochastic nature of the generative model exacerbates discrepancies between the input and output, especially in structurally complex regions.

To address these challenges, we propose the **Fidelity-Oriented Diffusion (FOD)**, a novel framework designed to enhance fidelity in diffusion-based super-resolution while preserving exceptional visual quality. Our approach introduces two key innovations: (1) a dual-memory module that strengthens conditional information through HQ feature guidance, and (2) an uncertainty-aware noise estimation strategy that prioritizes challenging reconstruction regions. Both of them are aimed at improving noise prediction accuracy. Specifically, the dual-memory module consists of an HQ memory bank to enhance the input features and a degradation memory to remove the degradation components from the low-quality features. The HQ memory bank is pre-trained on high-resolution images to store a rich set of high-quality features, capturing fine structures and textures. Using a cross-attention mechanism, we retrieve the corresponding HQ features for enhancing the LQ conditional information, ensuring that the generative process is guided by more detailed and reliable condition features. Furthermore, we develop an uncertainty-aware noise estimation strategy to better handle regions that are challenging to reconstruct. We add an extra branch in the denoising network to predict uncertainty values, identifying areas where distinguishing noise from the signal is difficult. Based on this estimated uncertainty, we design a weighted loss function that adaptively assigns greater importance to high-uncertainty regions, ensuring the model focuses more on area with strong noise-signal coupling. Our contributions are as follows:

- We identify and analyze the fundamental limitation in existing diffusion-based SR methods - the inaccuracy of predicted noise is due to low-quality conditional feature, which critically impacts image reconstruction fidelity.
- We propose a dual-memory module to enhance the conditional feature, which includes a high-quality memory bank to enrich the structural information and a degradation memory to remove the degradation components, thereby significantly reinforcing the guidance for diffusion processes.
- We develop an uncertainty-aware noise estimation framework that can dynamically adjust the optimization weights for high-uncertainty regions during the denoising process, significantly improving the accuracy of noise estimation in challenging reconstruction areas.
- Experimental results demonstrate that our method not only improves the quantitative fidelity metrics of diffusion-based super-resolution, but also present superior visual quality across diverse scenarios.

2 Related Work

2.1 Real-World Image Super-Resolution

Real-world image SR aims to upscale LR images that have unknown real-world degradation. Conventional CNN-based methods [8, 9, 24, 49, 50] typically assume predefined degradation models, which struggle to address real-world degradations. Recently, more flexible

approaches have been developed [34, 41]. Real-ESRGAN [37] and BSRGAN [46] introduce comprehensive degradation pipelines by simulating complex noise, blur and other real-world artifacts to enhance model robustness. DASR [25] incorporates a specialized module for estimating different degradation types and designs a degradation-aware SR network. FeMaSR [6] employs VQGAN to match the LR features. Many studies [3, 12, 23, 29, 35, 39, 48, 48] employ adversarial learning to produce more realistic results. Despite these advancements, real-world SR remains a challenging task due to the complexity of real-world degradation processes.

2.2 Diffusion-Based Super-Resolution

Diffusion-based SR methods have attracted attention for their ability to generate highly realistic images. By employing powerful pre-trained text-to-image models such as Stable Diffusion [30], they treat the SR problem as a generative process. StableSR [33] exploits diffusion prior by fine-tuning a time-aware encoder conditioned on the LR input, and incorporates a controllable feature wrapping module to balance realism and fidelity. SeeSR [42] introduces semantic-aware modules to guide the diffusion process, focusing on critical semantic information to achieve excellent details and structures. PASD [43] employs a pixel-aware module and a degradation removal network to perceive local structures. Although these approaches can produce impressive textures and details, they still face challenges in preserving precise fidelity with the input image, especially in complex real-world scenarios.

2.3 Uncertainty Learning

Uncertainty learning has been widely applied to boost the reliability and robustness of various applications [5, 14, 17, 21, 36], including image segmentation, medical imaging and image restoration. In low-light image enhancement [18], an uncertainty-guided learning framework is used to improve performance by modeling and reducing uncertainties in complex and unpredictable scenes. Similarly, [10] presents a learning-based image reconstruction framework that quantifies epistemic and aleatoric uncertainties through deep unrolling Bayesian neural networks. UFPNet [11] introduces uncertainty learning into the normalizing flow process by quantifying the corresponding uncertainty. Self-feature distillation methods [44] integrate uncertainty modeling to produce high-quality features, thereby improving the performance of image classification tasks. Ctrl-U [45] leverages the uncertainty-aware reward modeling to enhance the semantic alignment in image generation tasks.

3 Methodology

Existing diffusion-based SR methods often struggle with fidelity issues, which is due to the low-quality conditional features causing significant errors during the backward denoising process. To improve the noise estimation accuracy and thus boost the output fidelity, we first propose a dual-memory module to enhance the quality of input conditional features. Then, we introduce an uncertainty-aware noise estimation strategy to dynamically adjust the weight of challenging regions during optimization. The overall framework of the proposed Fidelity-Oriented Diffusion (FOD) method for real-world super-resolution is illustrated in Figure 3.

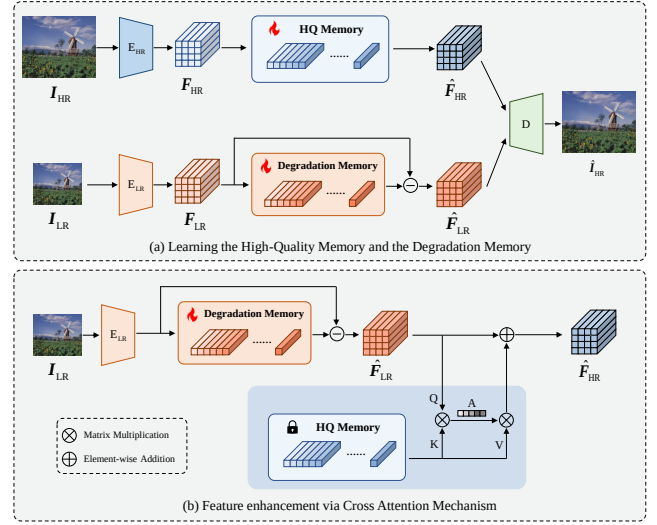


Figure 2: The architecture of the proposed dual-memory feature enhancement framework. (a) The high-quality features learned from HR images are stored in a dedicated HQ Memory bank, and the Degradation Memory captures real-world degradation patterns, we remove the degradation components from the LR features by a residual connection. (b) The LR features are enhanced by retrieving relevant high-quality features from the pre-trained HR memory bank via the cross-attention mechanism.

3.1 Memory-Enhanced Feature Refinement

In most diffusion SR frameworks, the input LR features are used as conditional information, which is low-quality and lacks essential high-frequency details to guide the generation of HR images. As a result, the restored features tend to deviate from the target features, affecting output fidelity. To address this issue, we propose a dual-memory module to enhance the input features, enriching the condition information and producing more faithful outputs.

3.1.1 Learning the HQ and Degradation Memory. To capture high-quality features, we first input diverse HR images I_{HR} into our encoder E_{HR} to extract deep features, as illustrated in the top of Figure 2 (a). These features are stored in a High-Quality (HQ) memory bank M_{HQ} , serving as a rich repository of high-quality information. We employ a self-supervised strategy [16] to update the memory bank $M \in \mathbb{R}^{n \times c}$, where n is the number of memory items. This strategy ensures that the memory bank dynamically adapts to the input features, while maintaining its relevance and accuracy over time. To achieve this, we first calculate the cosine similarity s_{ij} between the i -th memory item $e_i \in \mathbb{R}^{c \times 1 \times 1}$ and the j -th vector $f_j \in \mathbb{R}^{c \times 1 \times 1}$ of input image features F , which is defined as:

$$s_{ij} = \frac{e_i f_j^T}{\|e_i\| \|f_j\|}. \quad (1)$$

This similarity metric quantifies the alignment between the memory item and the input feature vector, ensuring that the most relevant information is retrieved. Subsequently, the index r of the most

relevant memory item e_r associated with the image feature vector f_j is determined by sorting out the highest similarity, i.e.,

$$r = \arg \max_i \{s_{ij}\}, \quad (2)$$

Then the retrieved memory item e_r is updated by its corresponding image feature vector through the rule $e_i \leftarrow \eta e_i + (1 - \eta) f_j$, where η is a decay rate. This self-supervised update mechanism enables the HQ memory bank to dynamically adapt to new HR input features, thereby preserving high-quality texture and structure information.

In parallel, the corresponding LR image I_{LR} is processed through an encoder E_{LR} to generate low-quality features, which are utilized to construct a degradation memory bank, as shown in the bottom of Figure 2 (a). The degradation memory bank serves as a repository for capturing real-world degradation patterns. By implicitly modeling these degradation patterns, the degradation memory bank provides a comprehensive understanding of the distortions in LR images, which is crucial for effective image restoration. To effectively construct the degradation memory bank, we employ a residual connection mechanism. The LR features are subtracted from the features of the degradation memory bank via a residual connection. This process aims to isolate and remove the degradation components, thereby enhancing the remaining features to achieve high-quality representations. Consequently, the residual connection ensures that the enhanced features are free from the distortions introduced by the degradation, leading to a more accurate and visually appealing reconstruction of the HR image.

To align the representations of two memory banks into a unified domain, both features F_{HR} and F_{LR} are decoded through the same decoder D , mapping the features back into the original HR image space. This alignment bridges the representation gap between the HQ and degradation memory banks, thereby improving the reconstruction capability of F_{LR} . However, the quality of F_{LR} still remains inferior to F_{HR} . To address this, we must enhance the quality of F_{LR} to obtain accurate conditional information, ensuring robust and high-fidelity image reconstruction.

3.1.2 Feature Enhancement via Memory Retrieval. As the update process is completed, the image feature can be read from the memory bank according to the similarity, which is used as the weights for the vector linear combination. Using the computed similarity, the weight of each memory item can be derived by a softmax operation:

$$w_{ij} = \frac{\exp(s_{ij})}{\sum_{i=1}^n \exp(s_{ij})}. \quad (3)$$

Then the memory-enhanced features are obtained by the weighted aggregation of memory items:

$$\hat{f}_j = \sum_i^n w_{ij} e_i \quad (4)$$

Through this retrieval mechanism, the memory bank dynamically selects the most relevant memory items to replace the original input features, as illustrated in Figure 2 (a). This dynamic retrieval enables the model to incorporate richer and more contextually appropriate information from the memory bank, significantly improving feature representation. Such enhancements are particularly critical for tasks like super-resolution, where capturing fine details and texture is essential for high-quality image reconstruction.

Building upon this memory retrieval framework, we further leverage the pre-trained HR memory to enhance low-quality features. As seen in Figure 2 (b), we employ a cross-attention mechanism to retrieve relevant and high-quality information from HR memory. For each query Q derived from the LR features, we compute its similarity with the keys K from the HQ memory:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad (5)$$

where Q and K are learned from F_{LR} and M_{HQ} by linear projections, respectively. Here, we adopt the original M_{HQ} features as V without projection to maintain the HR feature consistency. And d_k is the dimensionality of the keys. The cross-attention mechanism retrieves the most relevant high-quality features from M_{HR} to enhance the LR features, effectively bridging the gap between low-quality inputs and high-resolution outputs.

As illustrated in Figure 3 (a), the proposed dual-memory module can seamlessly integrate into the diffusion-based SR framework. This module enhances the input LR features, which are then fed into the ControlNet, producing refined conditional information for the subsequent SD model. By providing more detailed and richer guidance to the diffusion model, it becomes less affected by the inherent stochasticity, thereby reducing unwanted artifacts or hallucinations. With our dual-memory module, we effectively improve super-resolution performance and yield higher fidelity outputs that maintain better consistency with the original input.

3.2 Uncertainty-Aware Noise Estimation

Conventional diffusion-based SR methods, such as StableSR [42] et al, typically adopt Mean Squared Error (MSE) loss for training the denoising networks, which treat all spatial pixels equally with the assumption that the importance of all pixels is the same. However, considering that the texture and edge are inherently difficult to restore, these areas should be paid more attention to during the optimization process. This is consistent with our observation, as seen in Figure 1 (b), the predicted noise is heavily influenced by the input image in complex areas, leading to undesirable artifacts in the final output latent code. Inspired by the pioneering work [28], we propose an uncertainty-aware noise estimation strategy to improve the accuracy of the predicted noise, which dynamically adjusts the weight of each pixel based on the estimated uncertainty map, prioritizing areas where the model exhibits lower confidence or reconstructions are more challenging.

Learning Uncertainty for Diffusion Model. Following probabilistic modeling [21], we assume that the target noise pixel z_i follows a Normal distribution, parameterized by the predicted value $\hat{z}_i$ (mean) and its uncertainty θ (variance), formulated as

$$z_i = \hat{z}_i + \epsilon \theta_i, \quad (6)$$

where $\epsilon \in \mathcal{N}(0, I)$ denotes the standard Normal distribution. The commonly used MSE loss can be interpreted as a constant variance in Eq. (6), indicating that the variance across different pixels remains uniform. However, we should focus on regions that are harder or less confident to predict. Therefore, we propose to simultaneously estimate their uncertainty θ_i , and the likelihood function of Eq. (6)

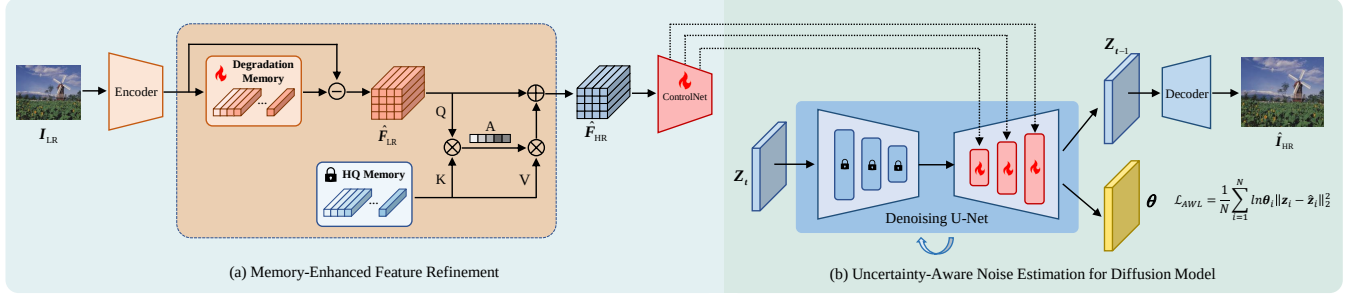


Figure 3: The overall framework of the proposed FOD method. (a) The dual-memory feature enhancement module is integrated to enhance the input LR features. (b) In the proposed uncertainty-aware noise estimation framework for the diffusion model, an extra branch is added to the denoising network to estimate the uncertainty map θ , which can be used to dynamically adjust the optimization weight of each pixel.

can be expressed by a Gaussian distribution:

$$p(z_i | \hat{z}_i, \theta_i) = \frac{1}{\sqrt{2\pi}\theta_i} \exp\left(-\frac{\|z_i - \hat{z}_i\|_2^2}{2\theta_i^2}\right), \quad (7)$$

where $\hat{z}_i$ denotes the predicted noise by the denoising network. Since the uncertainty θ_i is generally sparse, we employ a common sparse prior to model its prior term, expressed as $p(\theta) \propto \frac{1}{\theta^\alpha}$, where α is the scale coefficient. Then, the uncertainty learning process can be represented as the joint likelihood estimation with the prior:

$$\begin{aligned} p(z_i, \theta_i, \hat{z}_i) &= p(z_i | \hat{z}_i, \theta_i) p(\theta_i) \\ &= \frac{1}{\sqrt{2\pi}\theta_i} \exp\left(-\frac{\|z_i - \hat{z}_i\|_2^2}{2\theta_i^2}\right) \frac{1}{\theta_i^\alpha}. \end{aligned} \quad (8)$$

Taking the logarithm of Eq. (8) leads to the following log-likelihood:

$$\ln p(z_i, \theta_i, \hat{z}_i) \propto -\frac{\|z_i - \hat{z}_i\|_2^2}{\theta_i^2} - (2 + 2\alpha) \ln \theta_i. \quad (9)$$

To estimate the uncertainty θ_i equals to maximize the log-likelihood Eq. (9), which can be further translated as minimizing the following Uncertainty Learning (UL) loss:

$$\mathcal{L}_{UL} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{\theta_i^2} \|z_i - \hat{z}_i\|_2^2 + (2 + 2\alpha) \ln \theta_i \right), \quad (10)$$

where N is the total number of pixels in an image.

To achieve this, we add an extra estimation branch to the original denoising network for learning uncertainty in the diffusion model, as shown in Figure 3 (b). The uncertainty learning loss $\mathcal{L}_{UL}$ in Eq. (10) is used to constrain the noise estimation process, which comprises two terms:

(1) Data Term: the first term $\frac{1}{\theta_i^2} \|z_i - \hat{z}_i\|_2^2$, weights the ℓ_2 -norm error by the inverse of θ_i . When the Euclidean distance between the predicted noise $\hat{z}_i$ and the target noise z_i is large—indicating high reconstruction difficulty—the inverse term enforces the network to learn a larger θ_i for minimizing the loss. As a result, the learned θ_i provides an inherent measure of uncertainty across regions, reflecting the relative difficulty of restoring different image areas.

(2) Uncertainty Regularization Term: the second term $(2 + 2\alpha) \ln \theta_i$, serves as a regularizer to prevent the model from overestimating uncertainty to avoid penalization. This mechanism prevents the

model from collapsing into high-uncertainty predictions across all pixels, which could degrade reconstruction quality.

Adaptive Weighted Loss. With the estimated uncertainty θ_i , we propose an adaptive weighted loss that dynamically adjusts the optimization weight based on the uncertainty of each pixel. Unlike the uncertainty learning process, which uses the inverse of uncertainty to reduce the weight of high-uncertainty pixels, our goal here is to increase the weight of these pixels. This adjustment prioritizes difficult regions, enabling the model to focus on more challenging areas to reconstruct. The proposed Adaptive Weighted Loss ($\mathcal{L}_{AWL}$) is defined as:

$$\mathcal{L}_{AWL} = \frac{1}{N} \sum_{i=1}^N w_i \|z_i - \hat{z}_i\|_2^2, \quad (11)$$

where $w_i = \ln \theta_i$ is a monotonically increasing function derived from the uncertainty θ_i , ensuring numerical stability and preventing excessive weight increases.

The proposed adaptive weighted loss builds upon the uncertainty estimated in the denoising process. By dynamically assigning greater weight to high-uncertainty pixels, the diffusion model effectively emphasizes learning in the most challenging regions. As shown in Figure 1 (c), the noise predicted by our method is more accurate and closely resembles the standard Normal distribution, ultimately obtaining more precise latent code and improved reconstruction quality.

3.3 Training Framework

Our FOD method can be integrated into the diffusion-based SR framework, which consists of the following training pipeline:

Step 1: Pre-training the HQ and Degradation Memory. As illustrated in Figure 2 (a), we first pre-train the HQ and Degradation memory banks, which are learned through the self-supervised strategy. We only need to apply the $\mathcal{L}_1$ loss to constrain the reconstructed image, which is expressed as

$$\mathcal{L}_{\text{recon}} = \frac{1}{M} \sum_{i=1}^M \|I_{HR} - \hat{I}_{HR}\|_1, \quad (12)$$

where M is the total number of training samples. By providing diverse high-resolution images, we enable the encoder to extract

Methods	DIV2K Valid				RealSR				DRealSR			
	PSNR↑	SSIM↑	LPIPS↓	FID↓	PSNR↑	SSIM↑	LPIPS↓	FID↓	PSNR↑	SSIM↑	LPIPS↓	FID↓
BSRGAN [46]	24.58	0.6269	0.3351	44.22	26.39	0.7654	0.2670	141.28	28.75	0.8031	0.2883	155.63
Real-ESRGAN [37]	24.29	0.6372	0.3112	37.64	25.69	0.7616	0.2727	135.18	28.64	0.8053	0.2847	147.62
DASR [25]	24.47	0.6304	0.3543	49.16	27.02	0.7708	0.3151	132.63	29.77	0.8264	0.3126	155.58
FeMaSR [6]	23.06	0.5887	0.3126	35.87	25.07	0.7358	0.2942	141.05	26.90	0.7572	0.3169	157.78
LDM [30]	23.32	0.5762	0.3199	26.47	25.48	0.7148	0.3180	132.72	27.98	0.7453	0.3405	156.01
StableSR [33]	23.31	0.5652	0.3111	24.41	24.67	0.7055	0.3073	131.91	28.09	0.7493	0.3264	147.95
FOD-StableSR	23.63	0.5865	0.3245	27.41	25.25	0.7209	0.2950	136.66	28.42	0.7750	0.2998	152.46
DiffBIR [27]	23.17	0.5377	0.3669	31.83	24.84	0.6484	0.3650	133.60	25.95	0.6229	0.4668	185.60
FOD-DiffBIR	23.49	0.5612	0.3603	33.66	25.21	0.6825	0.3297	135.88	26.41	0.6703	0.4128	184.92
SeeSR [42]	23.71	0.5989	0.3193	25.17	25.18	0.7199	0.3000	126.11	28.17	0.7674	0.3173	147.87
FOD-SeeSR	24.29	0.6245	0.3121	24.95	26.04	0.7475	0.2793	123.29	28.80	0.7935	0.2938	144.29
PASD [43]	24.46	0.6210	0.3794	31.27	26.65	0.7659	0.2871	126.16	29.26	0.7957	0.3175	142.98
FOD-PASD	24.76	0.6288	0.3824	28.47	26.98	0.7699	0.2734	117.32	29.66	0.8094	0.3001	134.75

Table 1: The comparison results of our FOD method (indicated by a blue background) and other real-world SR methods on different test sets. We employ different diffusion-based methods as our backbone network to prove our effectiveness. For example, the FOD-PASD denotes using PASD as our backbone. Our method can significantly improve the fidelity metrics (such as PSNR and SSIM) compared to the original backbone method, while maintaining competitive perceptual metrics.

rich deep features stored in a dedicated HR memory bank. And capture the real-world degradation components in the Degradation Memory. This dual-memory retrieval mechanism is then integrated into the diffusion model for enhancing the LR input features.

Step 2: Applying the Uncertainty-Aware Noise Estimation.

We incorporate an extra branch into the denoising network to facilitate uncertainty estimation, as illustrated in Figure 3 (b). Then we can learn the pixel variance θ_i associated with the noise by Eq. (10). This variance effectively reflects the challenging regions to reconstruct. With the estimated uncertainty, we apply the adaptive weighted loss in Eq. (11), which dynamically adjusts the pixel weight based on the uncertainty map, ensuring that pixels with higher uncertainty receive more attention during training. The total Uncertainty-Aware loss is expressed as:

$$\mathcal{L}_{UA} = \mathcal{L}_{UL} + \mathcal{L}_{AWL} \quad (13)$$

This Uncertainty-Aware loss is used to replace the original MSE loss, and significantly enhances the model’s ability to reconstruct challenging areas, ultimately leading to better fidelity and superior reconstruction of the final outputs.

4 Experiments

4.1 Experimental Settings

Datasets. Following previous work [33], we train our method on several common datasets, including DIV2K [1], DIV8K [13], Flickr2K [31], OutdoorSceneTraining [38] and FFHQ [19]. The LR images are generated through the degradation pipeline of Real-ESRGAN [37]. For the test datasets, we evaluate on both synthetic and real-world datasets provided by [33]. The synthetic data consists of 3,000 randomly cropped LR-HR pairs (the HR resolution is 512×512) from the DIV2K validation set, generated using the same degradation pipeline as in training. For real-world testsets, the RealSR [4] and DRealSR [40] datasets are used.

Compared Methods. We compare our FOD method with existing state-of-the-art real-world SR methods, including GAN-based methods BSRGAN [46], Real-ESRGAN [37], DASR [25] and FeMaSR [6]. We use different diffusion methods as our backbone network, including StableSR [33], SeeSR [42], DiffBIR [27], PASD [43].

Evaluation Metrics. We measure fidelity metrics using PSNR and SSIM. Additionally, we use LPIPS [47] as a reference-based metric to reflect perceptual quality. To evaluate the distribution distance between LR and SR images, we utilize the FID [15] metric. For more metrics such as DISTS [7], MUSIQ [20] and CLIPQA [32], we put the results in supplementary materials.

Implementation Details. We adopt $\alpha = 1$ in Eq. (10). The size of HQ and Degradation memory bank is set to 512, which is sufficient for feature space representation. For various comparison backbones, we directly follow their official codes for implementation.

4.2 Comparison Results

Quantitative Comparison. To verify the effectiveness of our FOD method, we have employed different diffusion methods as our backbone network. As shown in Table 1, our FOD method demonstrates significant improvements in both PSNR and SSIM compared to the original backbone, indicating that the fidelity of these SD-based methods are greatly improved. Moreover, our method maintains competitive perceptual metrics like LPIPS and FID, although some do not reflect as much improvement. This disparity is because the focus on high fidelity may limit the gains in perceptual quality due to different optimization targets. Fidelity metrics emphasize structural accuracy and focus on reducing pixel-wise errors, leading to a precise and detailed reconstruction. However, perceptual metrics prioritize how natural and visually appealing the images look to the human eye, measuring the subjective quality of the output rather than strict pixel-level accuracy. As for the computational complexity, our method only adds a few convolutional layers and

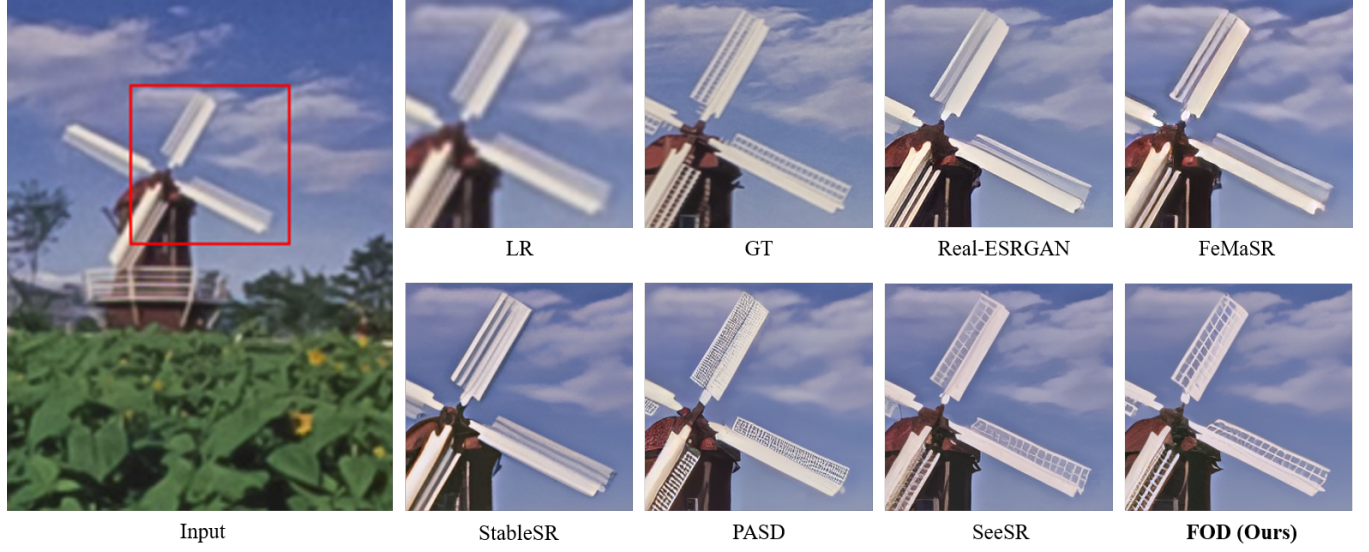


Figure 4: Visual comparisons results of $\times 4$ SR on DIV2K Valid datasets. From left to right: the input image, the partial zoom in LR image, the results of Real-ESRGAN [37], FeMaSR [6], StableSR [33], PASD [43], SeeSR [42] and FOD (Ours). Our method significantly reduces the hallucinations and artifacts present compared to other results.

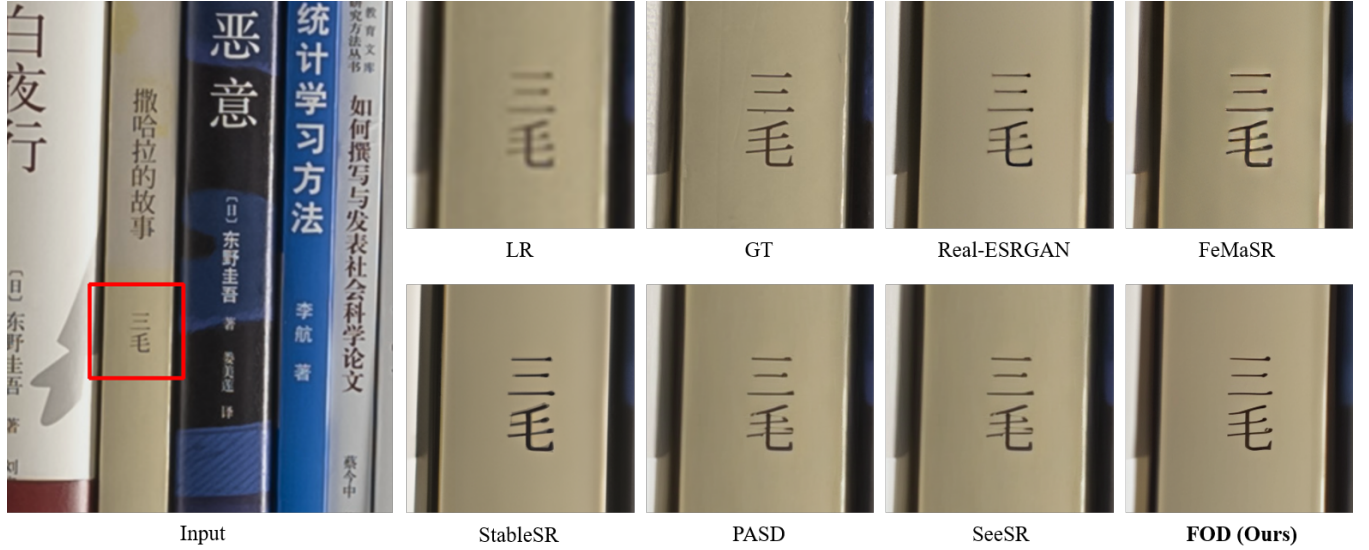


Figure 5: Visual comparisons results of $\times 4$ SR on RealSR datasets. Compared with other methods, our method can generate visually pleasant images that are closest to the real text strokes.

lightweight memory blocks to the backbone diffusion method, the increased parameter and FLOPs is 65.79K and 314.56M (on 128×128 image size) respectively, the computational complexity is less than 0.02% burden of the baseline model and can be almost negligible.

Visual Comparison. In addition to the quantitative improvements, our method also achieves superior perceptual quality by enhancing the conditional information and focusing on high-uncertainty regions. As illustrated in Figure 4 and 5, our FOD-SeeSR achieves three key advantages over conventional diffusion-based approaches:

- (1) significantly sharper structural edges, including the text characters;
- (2) More faithful texture reproduction; and
- (3) substantially reduced artifacts. These improvements are particularly noticeable in severely degraded areas, demonstrating our approach’s capability to maintain both high visual quality and ground-truth fidelity.

4.3 Ablation Studies

We evaluate the contribution of each component in our method using SeeSR [42] as the baseline model to get a clearer understanding

Models	M_D	M_{HQ}	$\mathcal{L}_{UA}$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
Baseline				28.17	0.7674	0.3173	147.87
A	✓			28.19	0.7689	0.3162	161.95
B		✓		28.21	0.7692	0.3288	160.98
C	✓	✓		28.44	0.7865	0.3095	156.67
D			✓	28.56	0.7879	0.3084	149.12
E	✓		✓	28.59	0.7882	0.3079	149.03
F		✓	✓	28.62	0.7885	0.3166	160.05
Ours	✓	✓	✓	28.73	0.7911	0.3042	147.41

Table 2: The ablation studies of different implementation models, our complete model performs best.

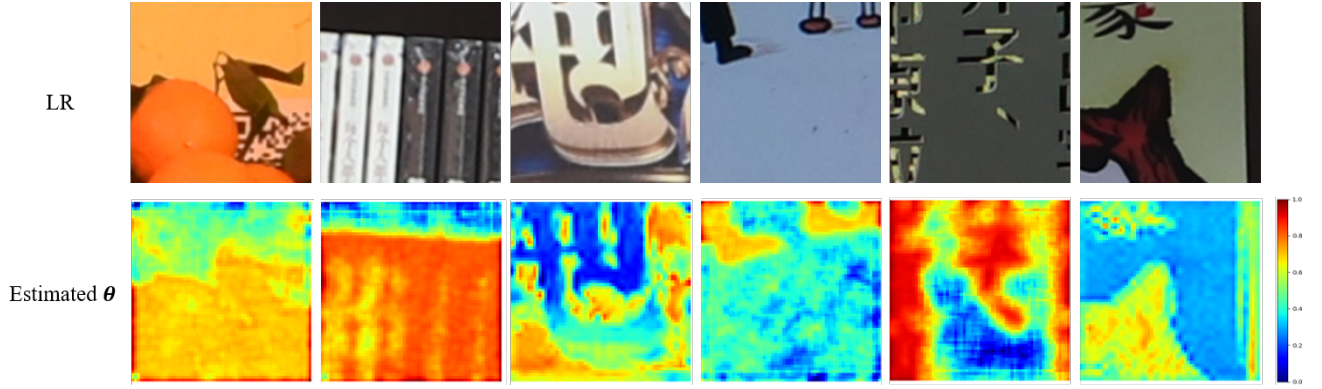


Figure 6: Visualizations of the estimated uncertainty maps θ (with normalization). Regions with higher reconstruction difficulty are assigned higher θ values, effectively reflecting the regions where noise prediction is particularly difficult.

of how each part improves the results. As shown in Table 2, our ablation studies compare the effects of HQ memory (M_{HQ}) and Degradation Memory (M_D), as well as the impact of the Uncertainty-Aware loss ($\mathcal{L}_{UA}$) on both fidelity and perceptual quality.

We compare the effects of using a single memory (only M_{HQ} or M_D) against the proposed dual-memory (both M_{HQ} and M_D) mechanism. As shown in Table 2, the dual-memory presents better image fidelity compared to single memory. This improvement is achieved by enhancing the conditional features with more diverse references, thereby reducing the risk of generating details that deviate from the true image structure. It can also be seen that incorporating the proposed uncertainty-aware loss into the base model significantly improves fidelity metrics. As presented in Figure 6, the estimated θ effectively measures the reconstruction difficulty, which allows the denoising network to dynamically adjust optimization weights, prioritizing regions that are challenging to reconstruct. This improvement demonstrates that the proposed uncertainty-aware loss is more effective than the traditional MSE loss in areas where detail preservation is crucial.

Our complete model (i.e., the baseline model with the dual-memory module and the uncertainty-aware loss) has achieved the best results, as seen in the last row of Table 2. This implementation leverages the advantages of feature enhancement (from the dual-memory mechanism) and uncertainty-aware noise estimation (from the $\mathcal{L}_{UA}$), effectively reducing unwanted artifacts and achieving

accurate reconstructions. This combination produces the highest fidelity metrics while preserving better perceptual results.

5 Conclusions

We propose the Fidelity-Oriented Diffusion (FOD), a novel framework that aims to improve the output fidelity while preserving visual quality for diffusion-based real-world super-resolution. Our analysis reveals that the primary source of reconstruction error is due to low-quality conditional features, which cause inaccurate noise prediction by the denoising network. To address this issue, our approach focus on improving the accuracy of noise estimation. First, we introduce a dual-memory module, which includes a high-quality memory bank to enhance the conditional features and a degradation memory to remove the degradation components. Relevant HQ features are retrieved and utilized through a cross-attention mechanism to enrich the conditional information. Furthermore, we develop an uncertainty-aware noise estimation framework that dynamically adjusts the pixel-level optimization weights according to the estimated uncertainty, assigning greater weight to complex (high-uncertainty) regions. This approach enables the model to focus on areas where the reconstruction is difficult, significantly improving the accuracy of noise estimation in challenging areas. Experimental results demonstrate that our method significantly improves fidelity metrics for diffusion-based SR while maintaining superior visual quality.

Acknowledgments

This work was supported by Innovation Capability Support Program of Shaanxi (Program No.2022TD-23), the Natural Science Foundation of China under Grant 62302369, Grant 62472336 and Grant 61991451.

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